

HAN YANG

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EDUCATION

University of Pennsylvania

Exchange in School of Engineering and Applied Science

Aug. 2023 - Dec. 2023

PA, USA

- Cumulative GPA: **4.000**/ 4.000
- Advanced Courses:
Computer Graphics; Natural Language Processing; Applied Machine Learning

The Chinese University of Hong Kong (CUHK)

B.Sc. in Computer Science

Sep. 2021 - May. 2025

Hong Kong SAR, China

- Cumulative GPA: **3.803**/ 4.000
- TOEFL iBT: 106/120
- Honors and Awards:
Yasumoto International Exchange Scholarship (2023, top 1 in ongoing exchange students); Award for Outstanding Academic Performance (2023, 2024, top 5% in CS major); Professor Omar Wing Memorial Scholarship (2022, top 1 in CS major); Dean's List (2022-2024, top 10% in the department); Honors at Entrance Scholarship (2021, top 1 in the Province)

RESEARCH INTEREST

Embodied AI, Robot Learning, Multi-modal Learning.

PUBLICATIONS & MANUSCRIPTS

Yuncong Yang*, **Han Yang***, Jiachen Zhou, Peihao Chen, Hongxin Zhang, Yilun Du, and Chuang Gan. 3D-Mem: 3D Scene Memory for Embodied Exploration and Reasoning. CVPR, 2025.

Han Yang, Kun Su, Yutong Zhang, Jiaben Chen, Kaizhi Qian, Gaowen Liu, and Chuang Gan. UniMuMo: Unified Text, Music, and Motion Generation. AAAI, 2025.

Han Yang*, Tianyu Wang*, Xiaowei Hu, and Chi-Wing Fu. SILT: A Shadow-aware Iterative Label Tuning Approach for Learning to Detect Shadows from Noisy Labels. ICCV, 2023.

RESEARCH EXPERIENCE

Lifelong scene memory for embodied exploration and reasoning

supervised by Prof. Chuang Gan, UMass Amherst & MIT-IBM Watson AI Lab

May 2024 - March 2025

MA, USA

- Proposed a snapshot-based scene representation that uses a set of informative images to represent scenes.
- Integrated with frontier-based exploration to facilitate active exploration with VLMs.
- Designed an incremental memory construction pipeline and a memory retrieval mechanism for lifelong memory.
- Achieved state-of-the-art performances on embodied question-answering and lifelong navigation benchmarks.

Unified generative model for music, motion and language

supervised by Prof. Chuang Gan, UMass Amherst & MIT-IBM Watson AI Lab

May 2023 - May 2024

MA, USA

- Able to perform conditional generation tasks on any combination of music, motion, and text.
- Adopted dynamic-time-warping to align unpaired music and motion and synthesize large-scale data.
- Employed a joint codebook for encoding music and motion and performed music-motion joint generation.
- Reduced computation demands by only requiring fine-tuning existing music and language models.

Shadow detection with self-training on noisy training data

supervised by Prof. Chi-Wing Fu, CUHK

May 2022 - Feb 2023

Hong Kong SAR, China

- Designed a self-training framework to train a shadow detection network and refine the noisy training data
- Proposed data augmentation methods to enhance the network’s understanding of shadow
- Relabeled and refined the test set of SBU-shadow dataset
- Surpassed previous state-of-the-art methods by large margins, *i.e.*, reduced Balanced Error Rate from 5.58 to 4.18.

EXTRA-CURRICULAR ACTIVITIES

- Won Honorable Mention Award in 2022 Mathematical Contest in Modeling.

SKILLS

Languages	Mandarin (native), English
Programming	Python, PyTorch, C/C++, OpenGL, Java, html, L ^A T _E X
Others	Figma, Pr, SolidWorks, Blender, TeleportHQ